Smart Home Monitoring System

Integrated IoT and CCTV Feed Analysis

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Presentation Outline

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Introduction to Smart Home Monitoring

- **Definition:** A system that allows users to monitor and control various home parameters remotely or automatically.
- **Core Components:**
  - **Sensors:** Temperature, humidity, motion, and light.
  - **Actuators:** Smart lights, thermostats, and electronic locks.
  - **Connectivity:** Wi-Fi, Bluetooth, or ZigBee.
- **The "Smart" Element:** Integration of data to provide a seamless user experience.

The Problem Statement

Current Challenges

- **Manual Monitoring:** Difficulty in keeping track of large homes simultaneously.
- **Security Gaps:** Lack of real-time alerts when away from home.
- **Energy Waste:** Lights and appliances left on in unoccupied rooms.

The Role of CCTV Feed

- **Visual Context:** While sensors tell you *that* something happened, CCTV shows you *what* happened.
- **Key Features:**
  - Real-time streaming.
  - Motion detection.
  - Night vision and Wide-angle views.
- **Integration:** Combining visual data with sensor data (e.g., a motion sensor triggers a specific CCTV zoom).

Key Features of the Proposed System

1. **Remote Accessibility:** Control the home via a smartphone/tablet.
2. **Real-time Alerts:** Instant notifications for motion, smoke, or temperature spikes.
3. **Live CCTV Feed:** A dedicated continuous stream for visual verification.
4. **Automation:** Scheduled tasks (e.g., "Turn on lights at 7 PM").

User Benefits

- **Security:** Peace of mind when traveling or sleeping.
- **Convenience:** Control everything with a single tap.
- **Energy Efficiency:** Reduced electricity bills through automated climate and light control.
- **Accessibility:** Easier navigation for elderly or physically challenged users.

Target Users

- **Homeowners:** For general comfort and aesthetics.
- **Security Conscious Individuals:** For 24/7 surveillance.
- **Elderly Care:** Monitoring the movement and health of seniors.
- **Commercial Use:** Small offices or retail shops.

Proposed Methodology Overview

The system is developed using a four-layer architecture:

1. **Perception Layer:** Sensors and CCTV Cameras.
2. **Network Layer:** Data transmission via Wi-Fi/MQTT protocols.
3. **Processing Layer:** Cloud computing or Edge processing (Raspberry Pi/Arduino).
4. **Application Layer:** Mobile App / Web Dashboard.

Hardware Selection

- **Microcontroller:** ESP32 or Raspberry Pi (chosen for built-in Wi-Fi).
- **Sensors:** DHT11 (Temp/Humidity), PIR (Motion), LDR (Light).
- **CCTV Module:** OV7670 or a high-definition IP Camera module.
- **Power Supply:** Li-ion batteries or AC adapters.

Software Stack

- **Programming Language:** Python / C++.
- **Communication Protocol:** MQTT (for low-latency sensor updates).
- **Database:** Firebase or MySQL to store historical logs.
- **Frontend:** Kotlin (Mobile) or HTML (Web).

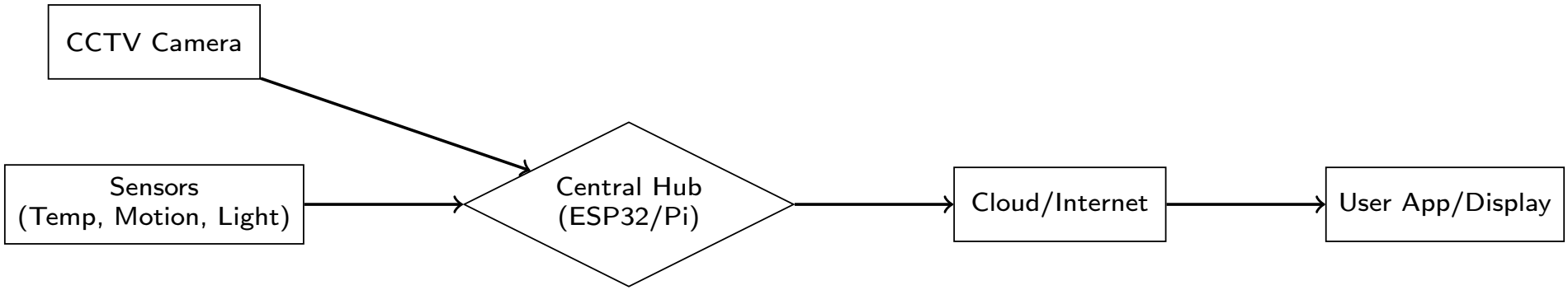
CCTV Processing Pipeline

- **Capture:** Raw video frames from the camera.
- **Preprocessing:** Grayscale conversion, resizing, and noise reduction.
- **Analysis:**
  - **Motion Detection:** Difference between consecutive frames.
  - **Object Detection:** Identifying people, pets, or vehicles.
- **Output:** Stream to the dashboard and trigger alerts.

Proposed Workflow

- **Step 1:** Initialize sensors and connect to the local Wi-Fi.
- **Step 2:** Continuous data streaming from sensors and CCTV.
- **Step 3:** Data is parsed and sent to the Cloud/Server.
- **Step 4:** User interacts with the dashboard to view data or toggle switches.
- **Step 5:** Feedback loop (e.g., User clicks "Light ON" $\rightarrow$ Actuator receives command $\rightarrow$ Light turns on).

System Block Diagram



Block Diagram Explanation (Part 1)

- **Sensors:** The primary input devices. They convert physical parameters (temperature, movement) into electrical signals.
- **CCTV Camera:** Provides a continuous visual stream. It acts as the "eyes" of the system.
- **Central Hub:** The brain of the system. It collects data from all inputs, processes the logic (e.g., "Is it dark?"), and sends the output.

Block Diagram Explanation (Part 2)

- **Cloud/Internet:** Acts as a bridge. It allows the user to monitor the home from a different location (Remote Monitoring).
- **User App/Display:** The interface where the user sees the live CCTV feed, current temperature, and toggles the smart lights.

Simulated Results (Visuals)

Placeholder for Result 1

(Insert Image/Graph of Sensor Data Trends here)

Simulated Results (Visuals)

Placeholder for Result 2

(Insert Screenshot of the CCTV Live Feed and Mobile Dashboard here)

References

- S. K. Singh et al., "IoT-enabled Smart Home Systems: A Survey," *Journal of Sensors*, 2020. *(Provides an overview of sensor integration and automation).*
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